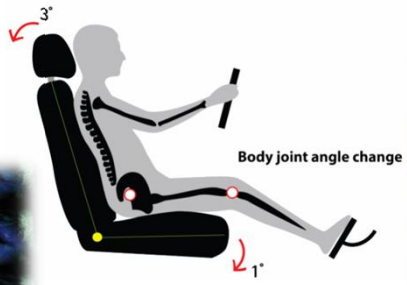
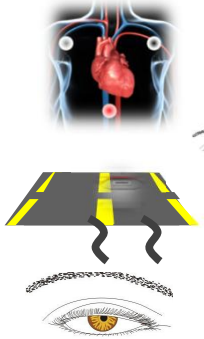
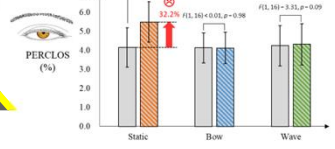
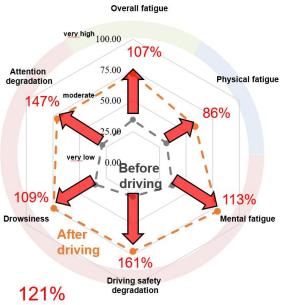
The Effects of a Motion Seat System on Driver's Passive Task-Related Fatigue: A Lab-Based Driving Simulation Study









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Types of Driving Fatigue

- Driver fatigue contributes to 20% to 30% of road collisions (Balkin et al., 2011)

Active task-related (TR) fatigue

- Increased task load
- High density traffic
- Poor visibility

Passive task-related (TR) fatigue

- Monotonous driving
- Extended driving
- Assisted driving (cruise control system)

Aggravated

Sleep-related (SR) fatigue

- Time of the day (circadian effects)
- Sleep deprivation
- Sleep restriction
- Sleep disorders

Driving performance ⇓

Risk of accident ⇑

[Source] May and Baldwin (2009). Driver fatigue: The importance of identifying causal factors of fatigue when considering detection and countermeasure technologies. *Transportation Research, 12*(3), 218-224.

Objectives of the Study

- Monotonous driving (e.g., uncrowded road) over an extended period
  - ✓ passive TR fatigue ⇑
  - ✓ drowsiness during driving ⇑, driving performance ⇓
  - ✓ possibility of car accident ⇑

Monotonous driving → Passive TR fatigue → Accident

Alertness ⇓
Brake reaction time ⇑

Inactive eye movement

Drowsy driving

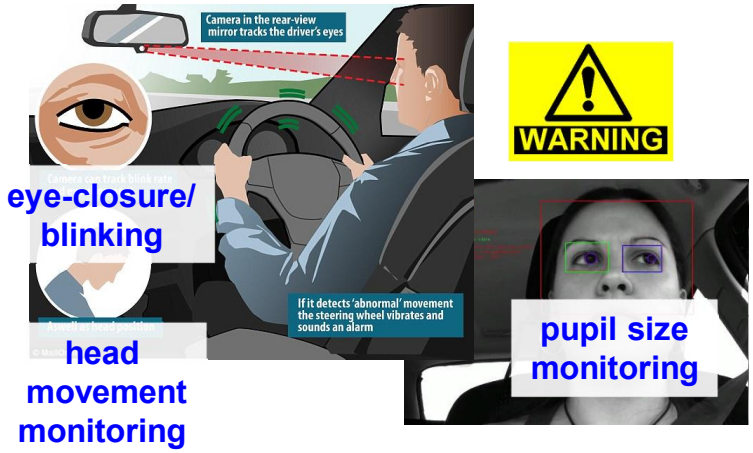
Limitations of Active Safety Systems

- **Collision avoidance system**: Prevents an accident using radar, laser (LIDAR), camera, and/or GPS sensors
- **Driver fatigue detection system**: Alerts the driver by seat vibration and/or audio alarm if a fatigue or distraction is detected by monitoring physiological signals (e.g., ECG), the driver's behaviors (e.g., eye-closure duration), and driving behaviors (e.g., lane departure)

⇒ **Assist and/or warn the driver** in driving environments, but **CANNOT prevent or mitigate the driver's fatigue**

Collision avoidance system

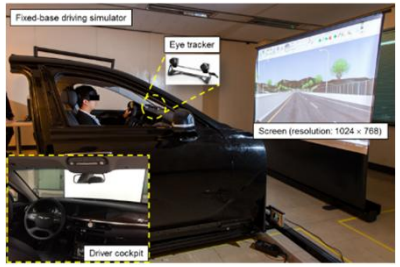
Drive fatigue detection system

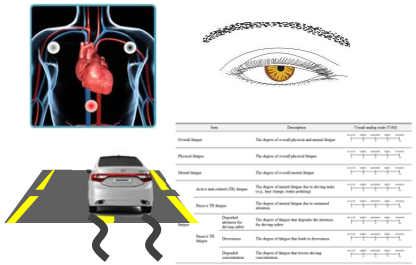


Objectives of the Study

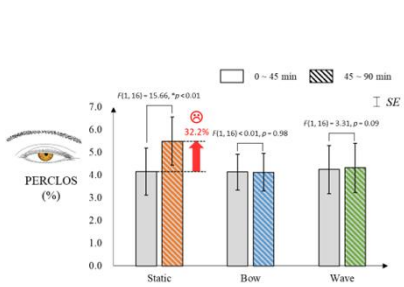
Evaluation of a **Motion Seat System**
for the Reduction of a **Driver's Passive Task-Related Fatigue**

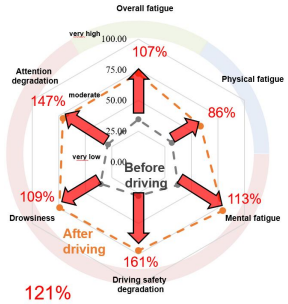
- Establish a **protocol to assess a driver's fatigue through simulated monotonous driving** in a lab environment





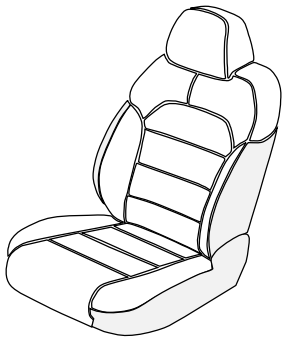
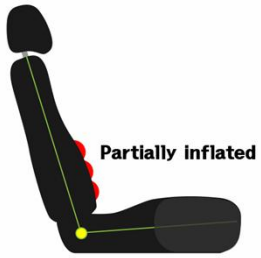
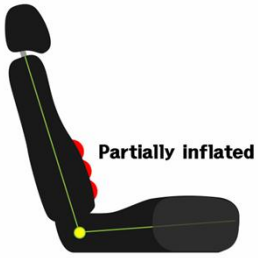
- Evaluate the **effects of a motion seat system on passive TR fatigue**



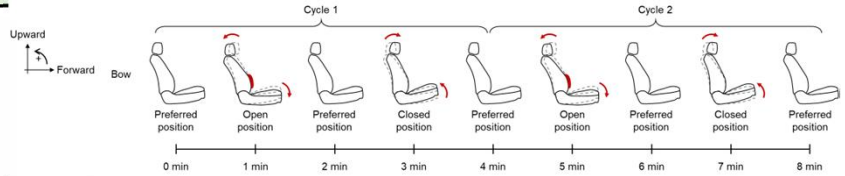


Motion Seat System: Motion Profiles

- Two types of seat motion profile were developed by combining the seatback angle, cushion angle, and lumbar support motions

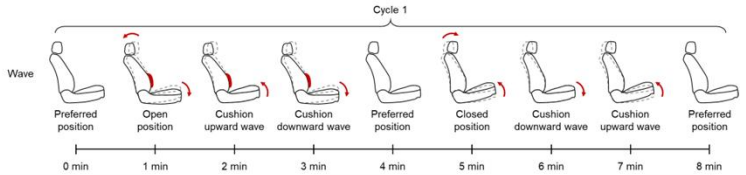
	Static seat (reference)	Motion seat profiles	
		Bow	Wave
		 Preferred posture (Neutral position)	 Preferred posture (Neutral position)
Seatback angle	Fixed	± 5°	
Cushion angle		± 5°	
Lumbar support		500 cc of air cell inflation/deflation	

Bow



Bow motion profile									
	Backrest	+5°	-5°	-5°	+5°	+5°	-5°	-5°	+5°
	Cushion	-5°	+5°	+5°	-5°	-5°	+5°	+5°	-5°
	Lumbar support	+500 cc	-500 cc			+500 cc	-500 cc		

Wave



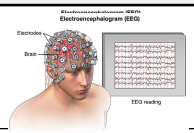
Wave motion profile									
	Backrest	+5°			-5°	-5°			+5°
	Cushion	-5°	+5°	-5°	+5°	+5°	-5°	+5°	-5°
	Lumbar support	+500 cc			-500 cc				

Fixed-Base Driving Simulator

- Reconfigurable seating buck: Fabricated using parts of EQ 900 (Hyundai-Kia Motors) and integrated with an ECU and the UC-win/Road ver. 10 (Forum 8, Japan)

Metrics of Passive TR Fatigue

- Used metrics that were found sensitive for the motion seat effect on passive TR fatigue in the pilot study (n = 6)

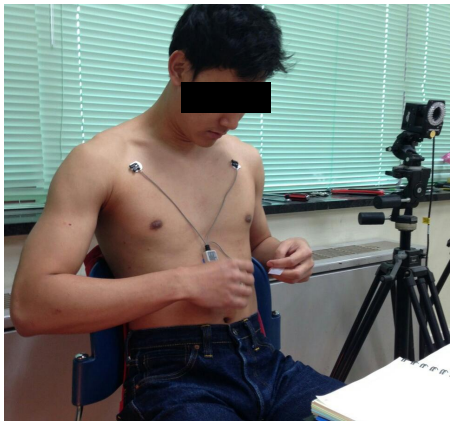
Passive TR fatigue assessment methods		Metrics	Tendency	References
Physiological Indicators	EEG 	α band β band δ band θ band	⇑ - - ⇑	Michail et al., 2008; Eoh et al., 2005; Jap et al., 2011; Yvonne et al., 2010; Cao et al., 2010;Zhao et al., 2011
	ECG	Heart rate (HR) NN interval (IBI) **SDNN** **LF/HF**	⇓ ⇓ ⇓ ⇑	Oron-Gilad et al., 2008; Ahsberg et al., 2000; Lai and Craig, 2001; Patel et al., 2011
	EDA (Skin conductance)	Tonic SCR	⇓	Lim et al., 1996
Psychomotor & Mental Tests	PVT (Psychomotor vigilance task)	Reaction time	⇑	Gunzelmann et al., 2011; Horrey and Wickens (2006)
Eye Movement	**Eye closure**	Amplitude Blinking rate Closing duration **PERCLOS**	⇓ ⇑ ⇑ ⇑	Lal and Craig, 2001b; Wijesuriya, et al., 2007; Svensson, 2004; Santamaria & Chiappa, 1987; Dinges et al., 1998;
Driving Performance	Steering wheel control	Angular position Steering wheel rate Yaw rate	⇑ ⇑ ⇑	Thiffault and Bergeron, 2010; He et al., 2011
	Longitudinal position control	SD of velocity **Brake reaction time**	⇑ ⇑	Pilutti and Ulsoy, 1999; Thiffault and Bergeron, 2010
	Lateral position control	**Lateral position** Lateral velocity Lateral acceleration	⇑ ⇑ ⇑	
Subjective Assessment	Karolinska sleepiness scale		⇑	Bekiaris et al., 2001
	Stanford Sleepiness Scale (SSS)		⇑	Hoddes et al., 1973
	Yoshitake Fatigue Scale		⇑	Yoshitake, 1978
	Dundee Stress State Questionnaire (DSSQ)		⇑	Matthews et al., 2002

ECG & Eye Tracker

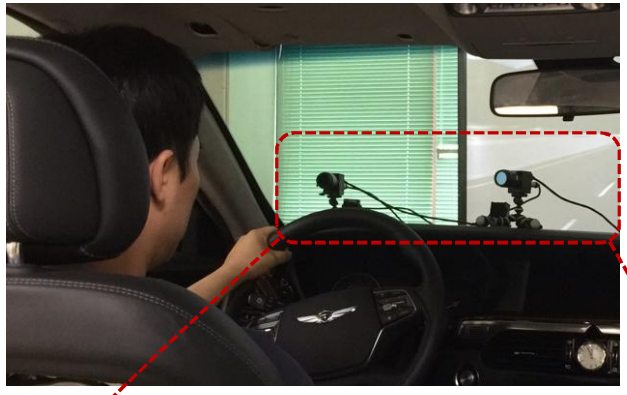
- Reconfigurable seating buck: Fabricated using parts of EQ 900 (Hyundai-Kia Motors) and integrated with an ECU and the UC-win/Road ver. 10 (Forum 8, Japan)

ECG measurement system

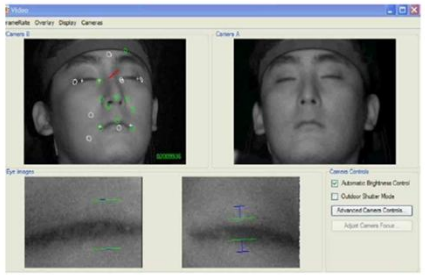
Noraxon Telemyo DTS
(Noraxon Inc., USA)



Eye tracking system

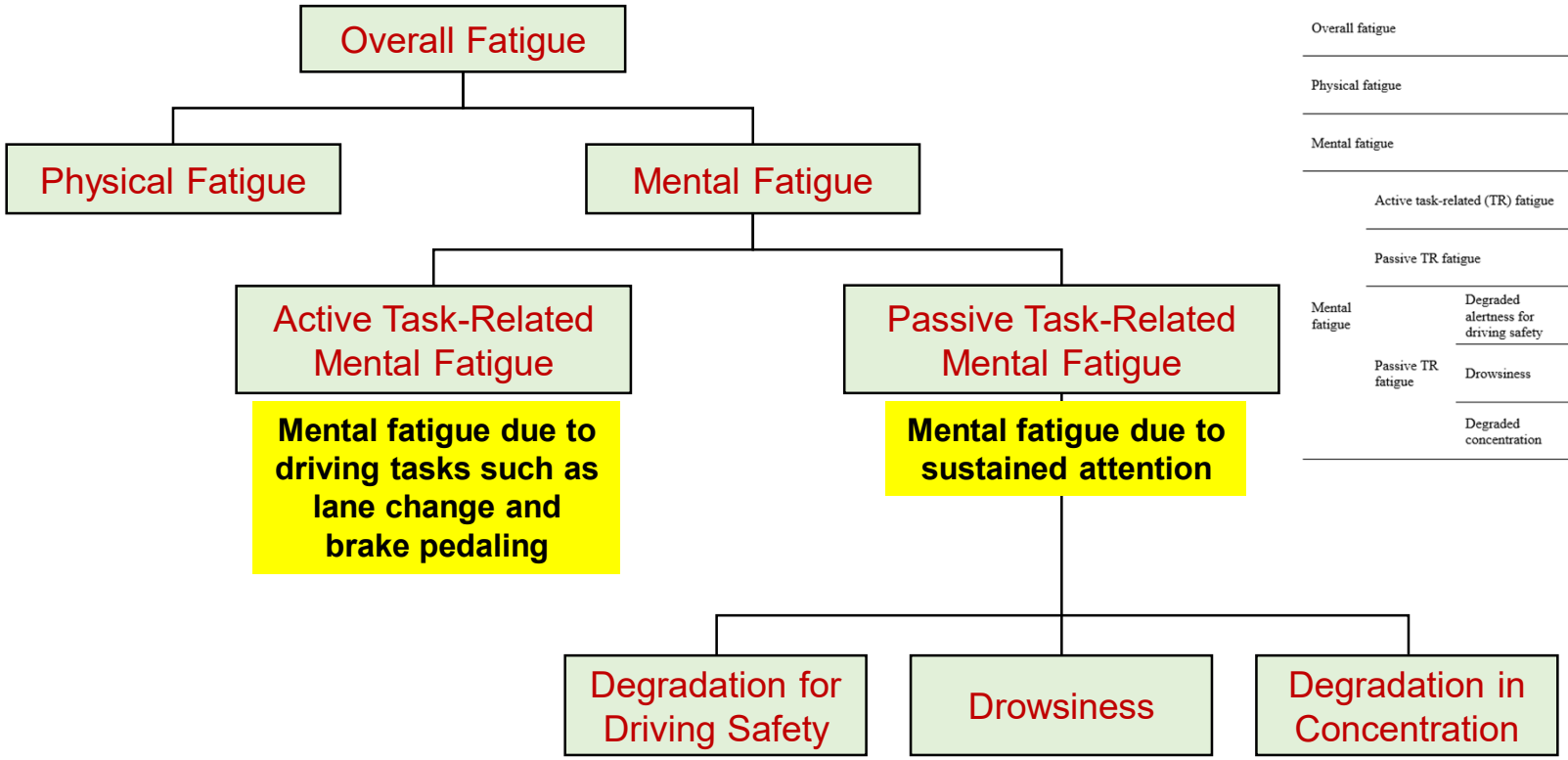


faceLAB 5
(Seeing Machines, Australia)



Subjective Fatigue: VAS

- Adopted the Dundee Stress State Questionnaire (DSSQ; Matthews et al., 2002) to assess the overall, physical, and mental fatigue before and after driving



Item			Description	Visual analog scale (VAS)
Overall fatigue			The degree of overall physical and mental fatigue	not at all slightly moderately highly extremely (0 25 50 75 100)
Physical fatigue			The degree of overall physical fatigue	not at all slightly moderately highly extremely (0 25 50 75 100)
Mental fatigue			The degree of overall mental fatigue	not at all slightly moderately highly extremely (0 25 50 75 100)
Mental fatigue	Active task-related (TR) fatigue		The degree of mental fatigue due to driving tasks (e.g., lane change, brake pedaling)	not at all slightly moderately highly extremely (0 25 50 75 100)
Mental fatigue	Passive TR fatigue		The degree of mental fatigue due to sustained attention	not at all slightly moderately highly extremely (0 25 50 75 100)
Mental fatigue	Passive TR fatigue	Degraded alertness for driving safety	The degree of fatigue that degrades the alertness for driving safety	not at all slightly moderately highly extremely (0 25 50 75 100)
Mental fatigue	Passive TR fatigue	Drowsiness	The degree of fatigue that leads to drowsiness	not at all slightly moderately highly extremely (0 25 50 75 100)
Mental fatigue	Passive TR fatigue	Degraded concentration	The degree of fatigue that lowers driving concentration	not at all slightly moderately highly extremely (0 25 50 75 100)

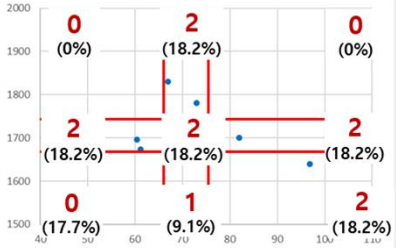
Participants

- 17 Korean drivers (M: 11, F: 6), age: 20s ~ 50s, driving experience: > 2 yrs., no history of musculoskeletal & cognitive disorders
- Representativeness of the participant group compared with Size Korea (2010) in terms of stature and weight

Dimension	Gender	Group	*n*	Mean	SD	Mean Comparison	SD Comparison
Stature (cm)	M	Size Korea	2471	171.4	6.1	$t(2480) = 0.21$, $p = 0.83$	$f(2470, 10) = 0.83$, $p = 0.28$
		Motion Seat	11	171.0	5.6		
	F	Size Korea	2025	158.4	5.6	$t(2480) = 0.73$, $p = 0.49$	$f(2470, 5) = 0.71$, $p = 0.22$
		Motion Seat	6	159.8	4.7		
Weight (kg)	M	Size Korea	2471	71.6	10.3	$t(2482) = -0.55$, $p = 0.60$	$f(2470, 10) = 1.04$, $p = 0.47$
		Motion Seat	11	73.3	10.5		
	F	Size Korea	2025	55.9	7.6	$t(2480) = -0.26$, $p = 0.81$	$f(2470, 5) = 0.71$, $p = 0.21$
		Motion Seat	6	55.2	6.4		

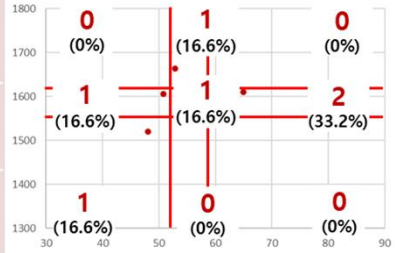
Male (n = 11)

stature \ weight		≤ 33rd	34th ~ 67th	≥ 67th
		≤ 66.5 kg	66.5 ~ 75.2 kg	≥ 75.2 kg
> 67th	≥ 174.3 cm	0 (0%)	2 (18.2%)	0 (0%)
34th ~ 67th	168.6 ~ 174.3 cm	2 (18.2%)	2 (18.2%)	2 (18.2%)
≤ 33rd	≤ 168.6 cm	0 (17.7%)	1 (9.1%)	2 (18.2%)



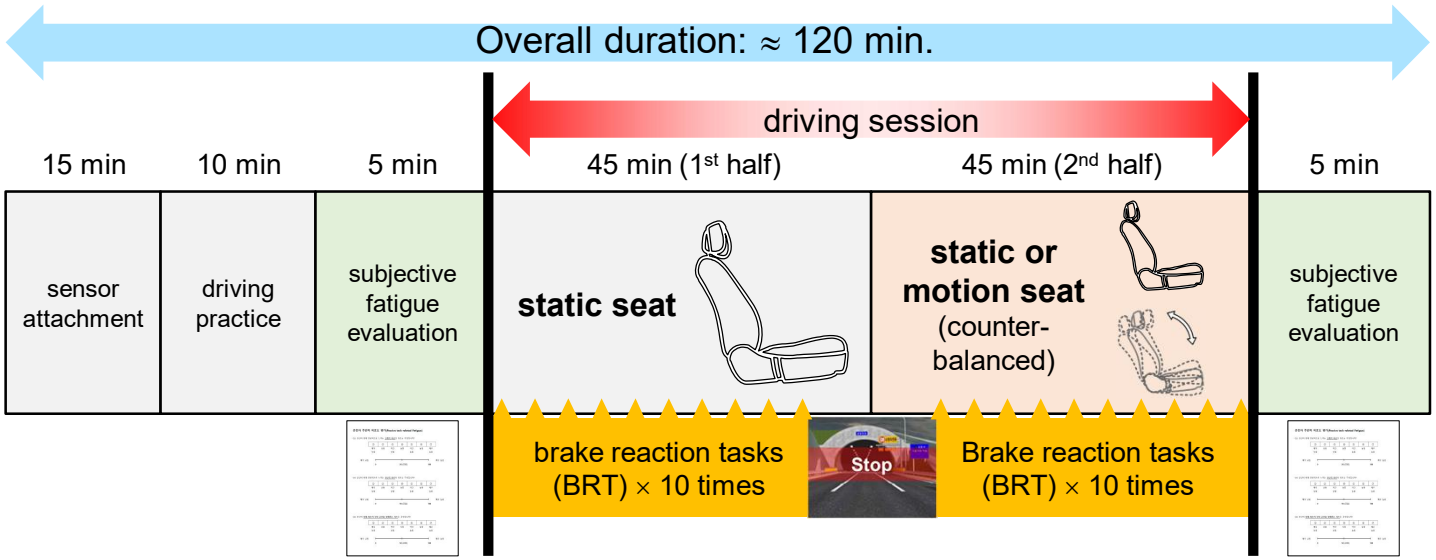
Female (n = 6)

stature \ weight		≤ 33rd	34th ~ 67th	≥ 67th
		≤ 52.0 kg	52.0 ~ 58.4 kg	≥ 58.4 kg
≥ 67th	≥ 161.0 cm	0 (0%)	1 (16.6%)	0 (0%)
34th ~ 67th	155.8 ~ 161.0 cm	1 (16.6%)	1 (16.6%)	2 (33.2%)
≤ 33rd	≤ 155.8 cm	1 (16.6%)	0 (0%)	0 (0%)



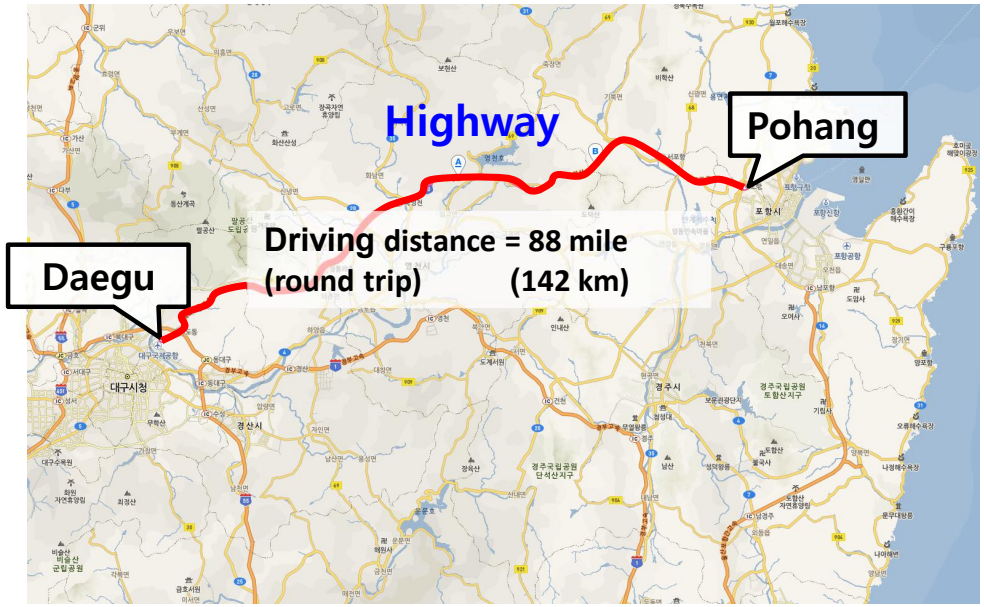
Experiment Procedure

- **Preparation (before driving 30 min)**: (1) attachment of sensors, (2) familiarization with driving in the simulator, (3) evaluation of subjective fatigue
- **Main experiment (driving; 90 min.)**: measurement of driving performance, eyelid closure, and ECG in static or motion seat conditions (counter-balanced)
- **Debriefing (after driving; 5 min)**: evaluation of subjective fatigue



Driving Environment

- Driving environments of **the highway of Daegu-Pohang** was **reconstructed using Korean map data and the driving simulation S/W UC-win/Road** ver. 10 (Forum8, Japan)
  - ✓ Driving distance (round trip) = 88 mile
  - ✓ Driving speed: 55 ~ 62 mph (90 ~ 100 km/h)

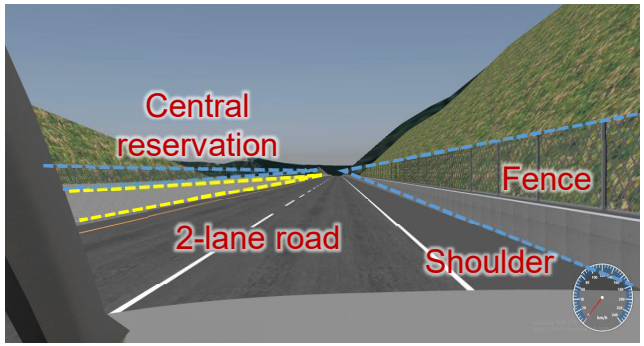


Reconstruction of Driving Environment

- Driving environments of the highway of Daegu-Pohang was reconstructed using Korean map data and the driving simulation S/W UC-win/Road ver. 10 (Forum8, Japan)

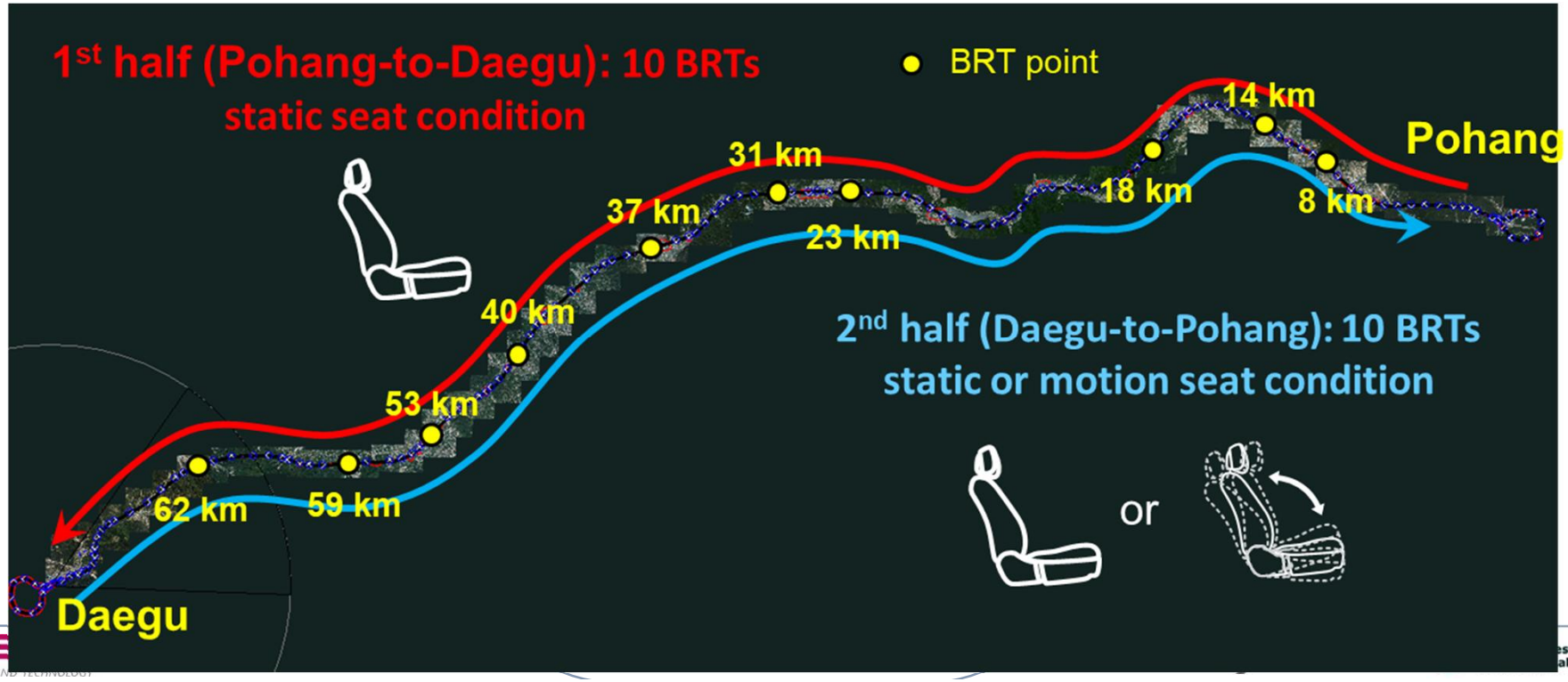
Real view

Reconstructed view



Brake Reaction Task (BRT)

- Measured a **break reaction time on a straight driving road area**
  - ✓ 1st half (Pohang-to-Daegu; static seat): 10 BRTs
  - ✓ 2nd half (Daegu-to-Pohang; static or motion seat):10 BRTs

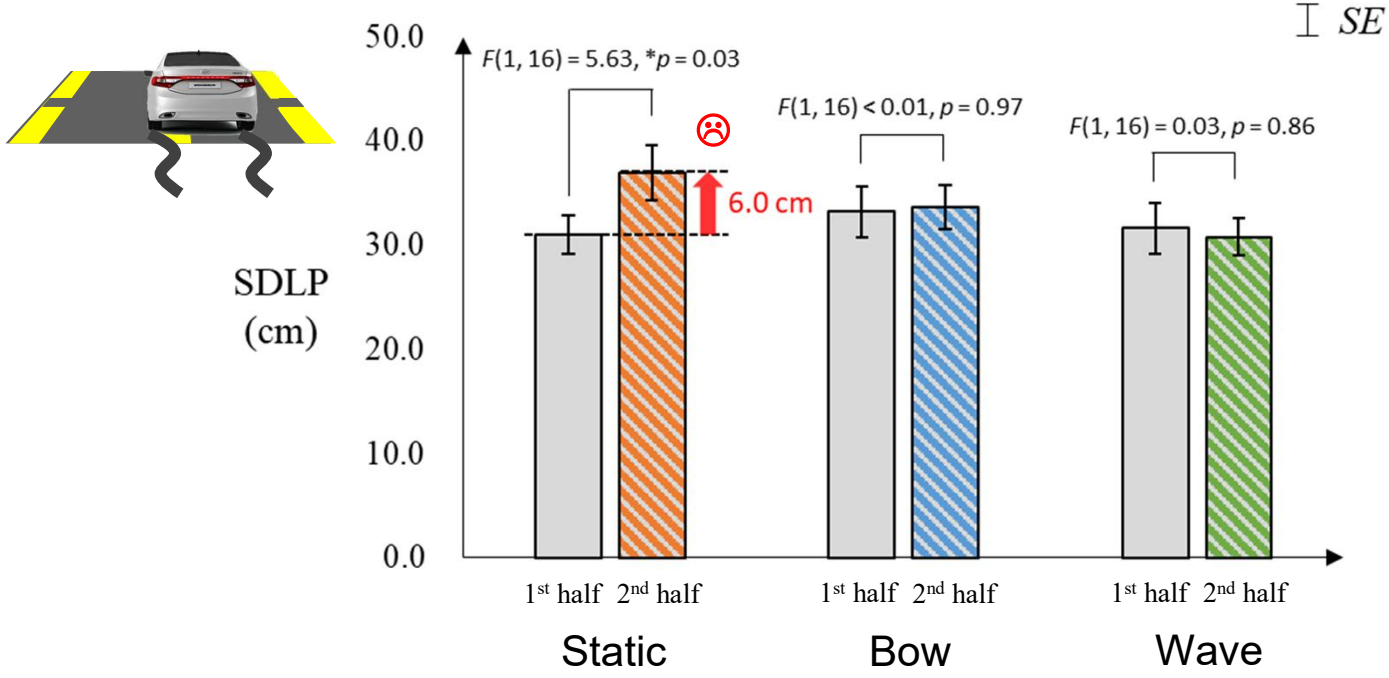


SD of Lane Position (SDLP)

- Static seat condition
  - ✓ 1st half: 31.0 ± 1.9 cm
  - ✓ 2nd half: 37.0 ± 2.6 cm

  } 6.0 cm (9.5%) ⇑ ⇒ Degraded driving performance

- Motion seat condition: No significant changes in SDLP ⇒ Maintained driving performance

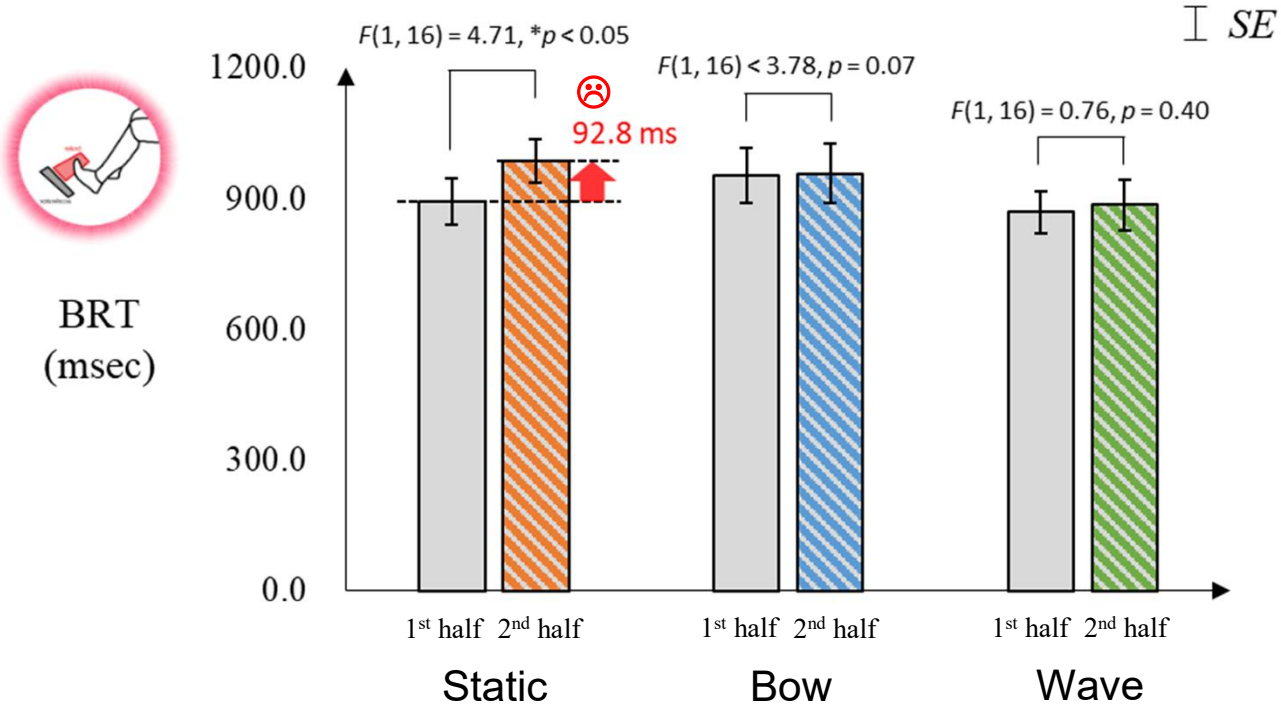


Brake Reaction Time (BRT)

- Static seat condition
  - ✓ 1st half: 893.6 ± 53.4 msec
  - ✓ 2nd half: 986.3 ± 50.5 msec

  } 92.8 ms (10.4%) ⇑ ⇒ Degraded driving performance

- Motion seat condition: No significant changes in BRT ⇒ Maintained driving performance

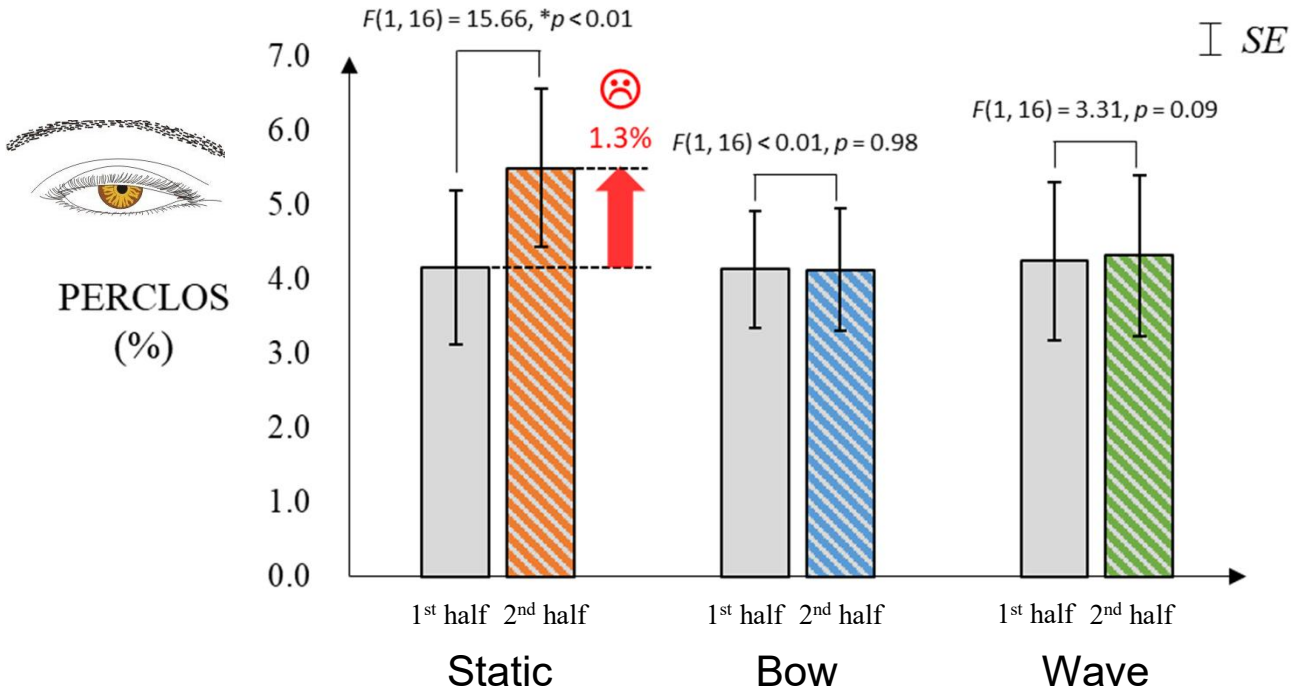


Percentage of Eye Closure (PERCLOS)

- Static seat condition
  - ✓ 1st half: 4.2% ± 1.0%
  - ✓ 2nd half: 5.5% ± 1.1%

  } 1.3% (32.2%) ⇑ ⇒ Increased driving fatigue

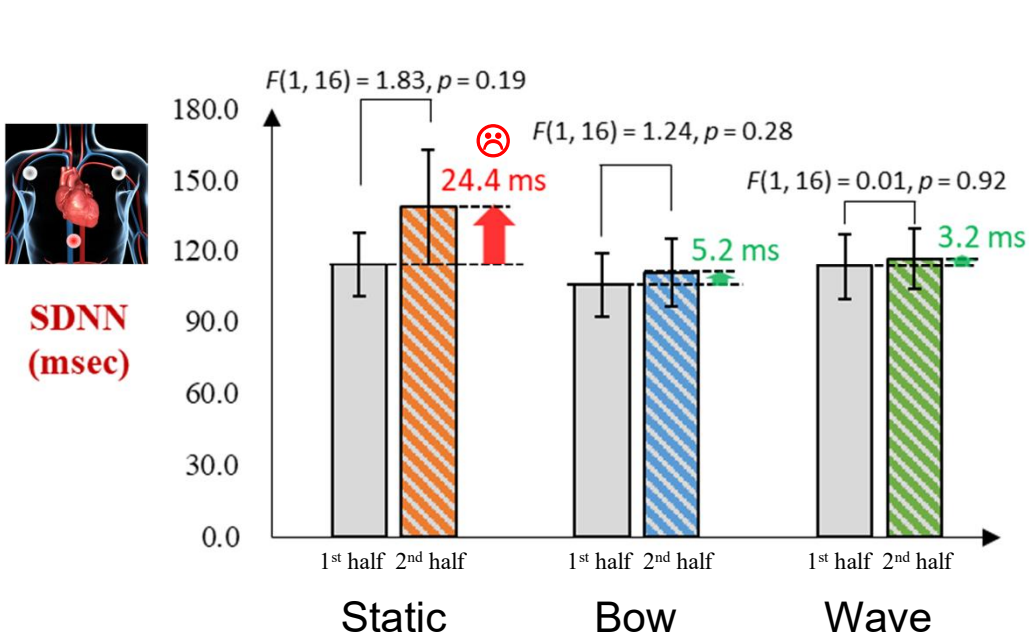
- Motion seat condition: No significant changes in PERCLOS ⇒ Maintained driving performance

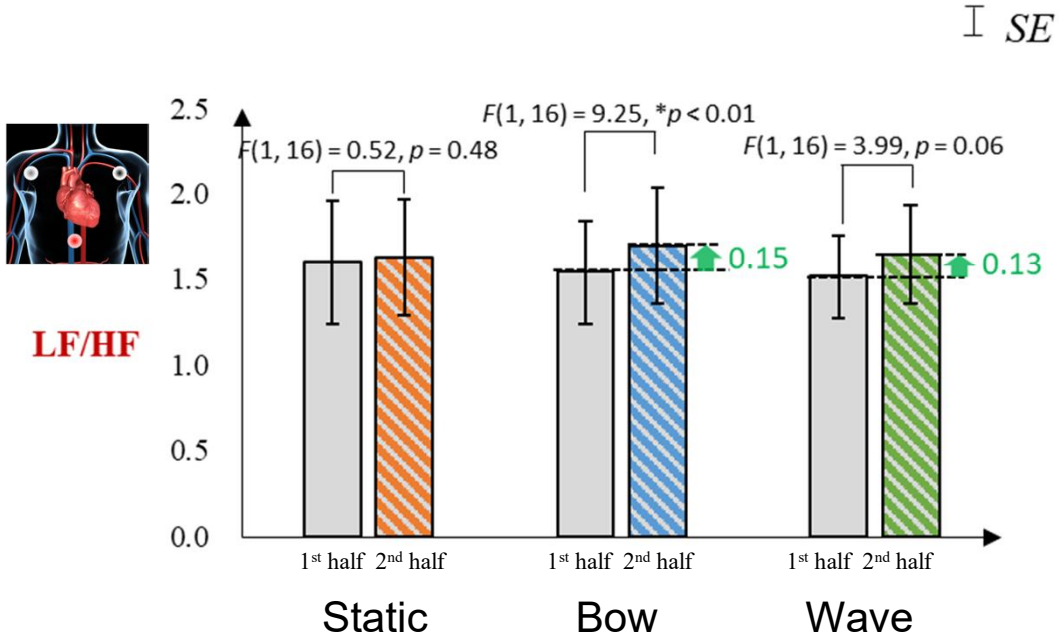


ECG HRV: SDNN & LF/HF

- Static seat condition
  - ✓ SDNN: 1st half vs. 2nd half ⇒ 1.3% (32.2%) ⇑ ⇒ Increased mental fatigue
  - ✓ LF/HF: No significant change ⇒ Maintained mental fatigue

- Motion seat condition: No significant changes in HRV ⇒ Maintained mental fatigue

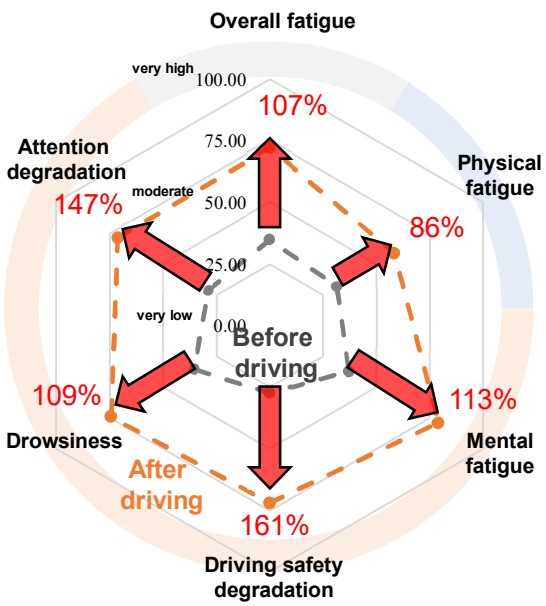




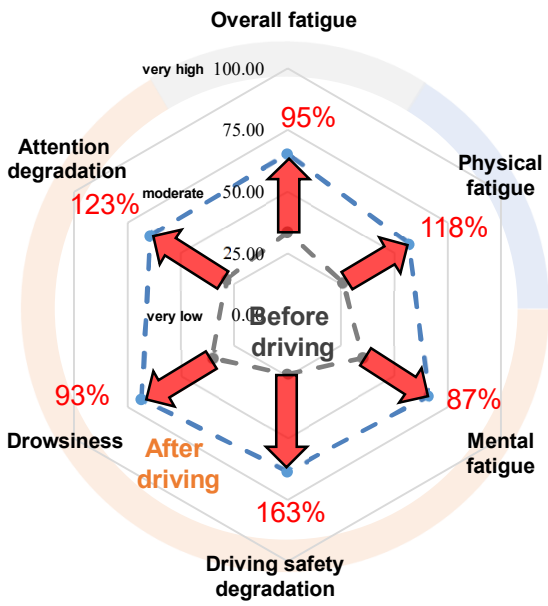
Subjective Fatigue: VAS

- Increased subjective fatigue for all seat conditions ($p < 0.05$)
- Δ subjective fatigue (%): **Static** (121% ☹) > **Bow** (113% 😐) > **Wave** (95% ☺)

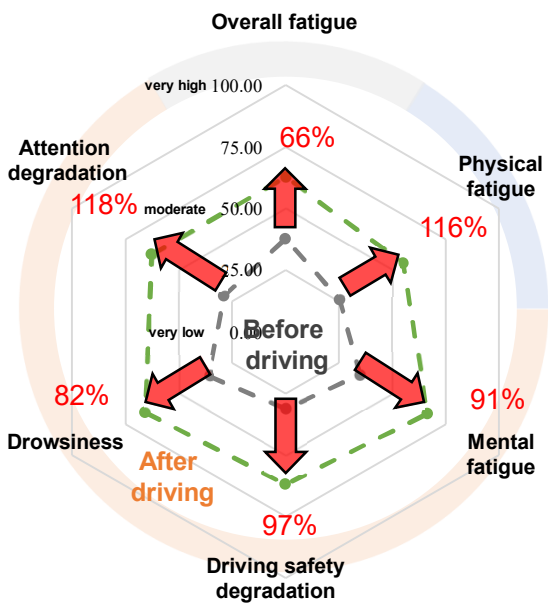
Static **121% ⇑**



Bow **113% ⇑**



Wave **95% ⇑**

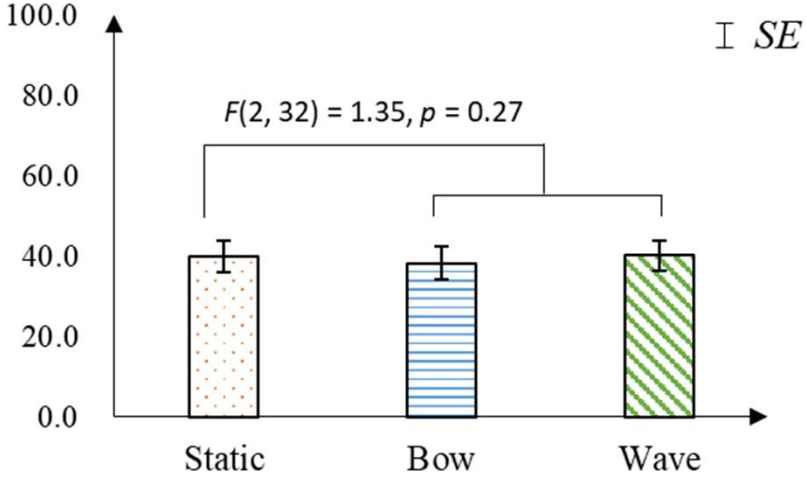


Active vs. Passive TR Fatigue: VAS

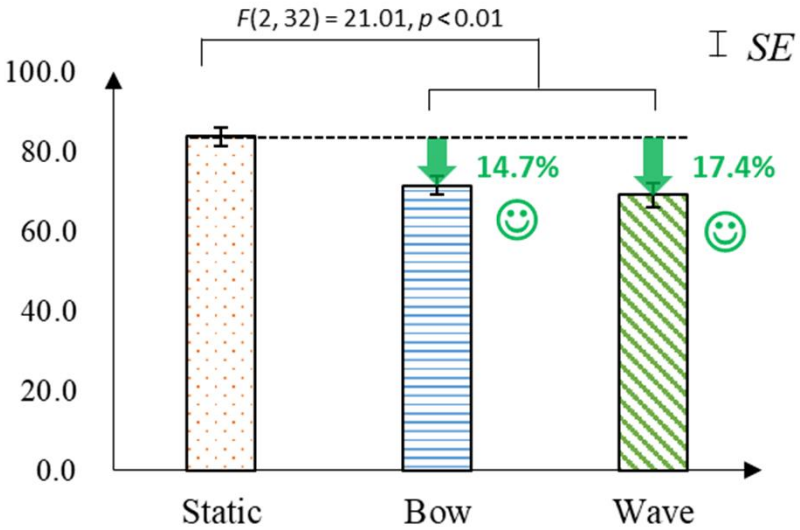
- **Active TR fatigue**: **Static** (40.1) ≈ **Bow** (38.3) ≈ **Wave** (40.2)
- **Passive TR fatigue**: **Static** (83.7 ☹) > **Bow** (71.4 ☺) ≈ **Wave** (69.1 ☺)

⇒ **Motion seat conditions (Bow, Wave)** were found **effective** to **reduce passive TR fatigue** in monotonous driving when compared with the static seat condition

Active TR fatigue



Passive TR fatigue



Summary: Static vs. Motion Seat Systems

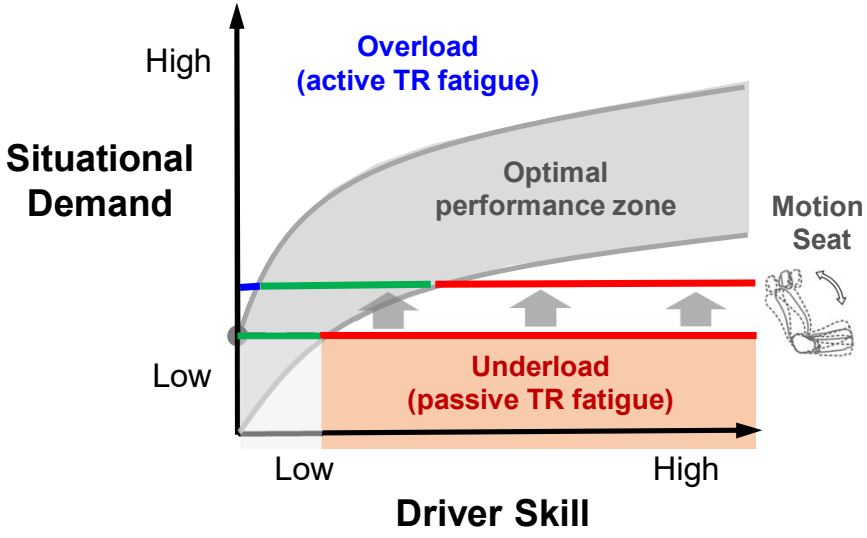
- The **motion seat conditions** (17% for Bow & 19% for Wave) **reduced passive TR fatigue** as compared with the static seat condition (33%)
- **Motion seat conditions** were found **reducing driving fatigue for objective metrics, but not for subjective metrics**

Metrics			Static				Bow				Wave			
			Static	Static	*MD*	Δ (%)	Static	Bow	*MD*	Δ (%)	Static	Wave	*MD*	Δ (%)
Driving perfor-mance	BRT (ms)		893.57	986.34	92.78 (*p<0.05)	⇑ (10.4%)	954.90	957.88	2.98	≈ (0.3%)	868.69	885.31	16.61	≈ (1.9%)
	SDLP (cm)		30.95	36.97	6.02 (*p=0.03)	⇑ (19.5%)	33.20	33.67	0.47	≈ (1.4%)	31.61	30.77	-0.84	≈ (-2.7%)
Mental fatigue	ECG	SDNN (ms)	114.15	138.58	24.44	≈ (21.4%)	105.58	110.75	5.17	≈ (4.9%)	113.43	116.64	3.21	≈ (2.8%)
		LF/HF	1.60	1.63	0.03	≈ (1.7%)	1.54	1.70	0.15 (*p<0.01)	⇓ (10.0%)	1.51	1.65	0.13	≈ (8.7%)
	PERCLOS (%)		4.16	5.49	1.34 (*p<0.01)	⇑ (32.2%)	4.13	4.12	-0.01	≈ (-0.3%)	4.25	4.31	0.07	≈ (1.6%)
Sub-jective fatigue	VAS		37.18	79.12	41.94 (*p<0.01)	⇑ (112.8%)	35.35	65.94	30.59 (*p<0.01)	⇑ (86.5%)	34.82	66.47	31.65 (*p<0.01)	⇑ (90.9%)
Overall			**33.0% ⇑**				**17.1% ⇑**				**18.9% ⇑**			

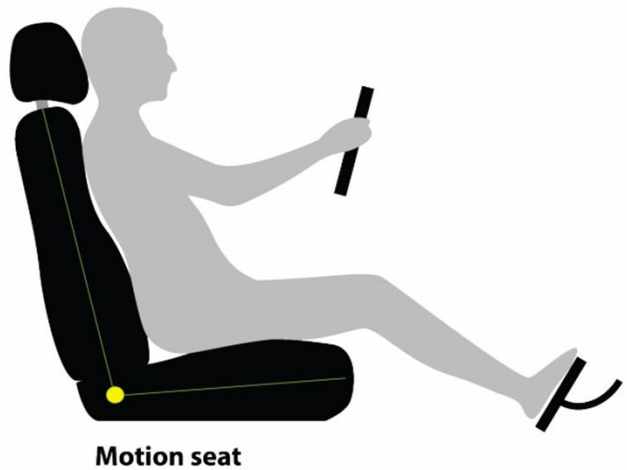
Motion Seat on Passive TR Fatigue Reduction

- Hypothetical mechanism of motion seat effects on passive TR fatigue reduction
  - ✓ The level of workload = *f* {situational demand, driver skill}
  - ✓ Motion seat activation ⇒ posture change ⇒ body joint angle change ⇒ proprioceptor activation ⇒ parietal lobe activation ⇒ situational demand ⇑ ⇒ passive TR fatigue ⇓

Situational Demand vs. Driver Skill
(Oron-Gilad et al., 2008)

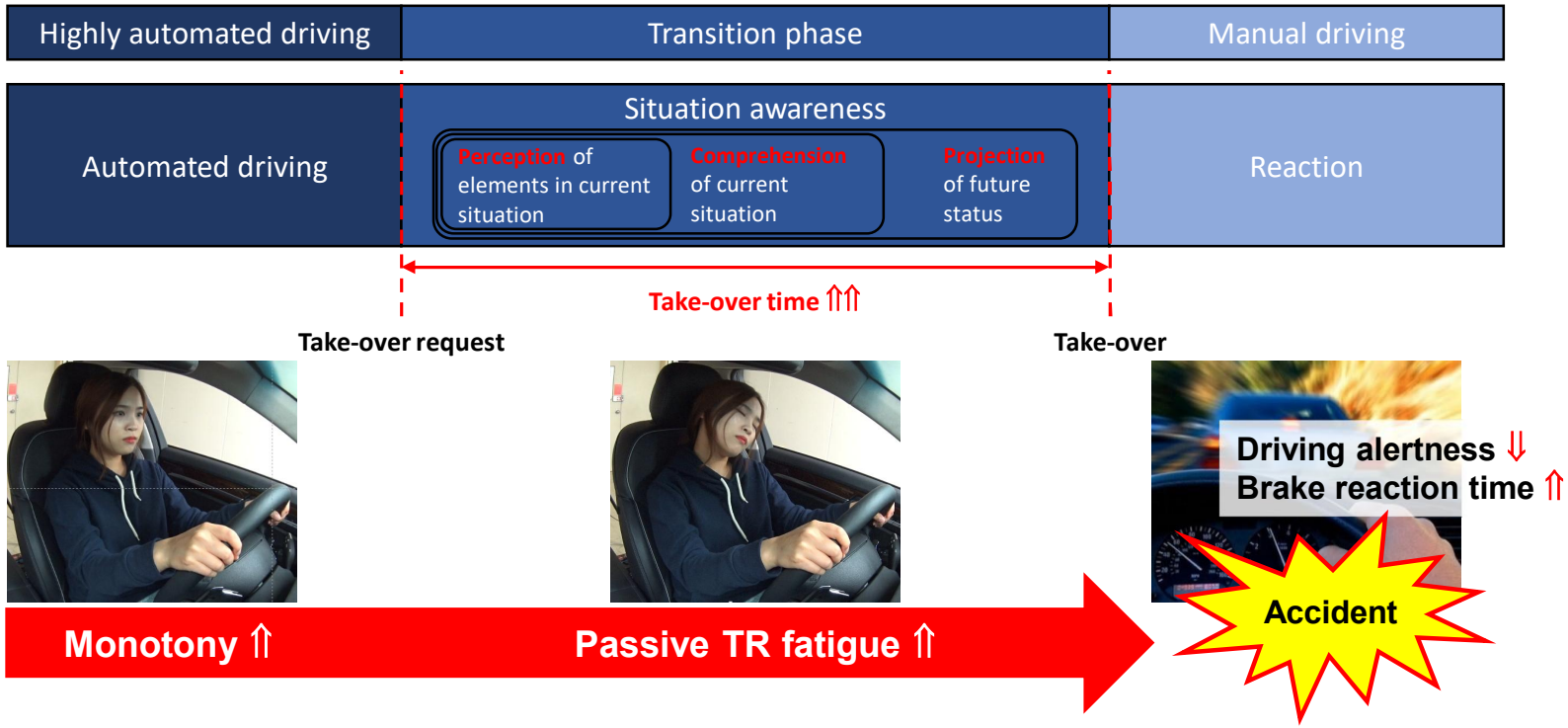


Hypothetical Mechanism of Motion Seat Effects



Motion Seat in Autonomous Driving

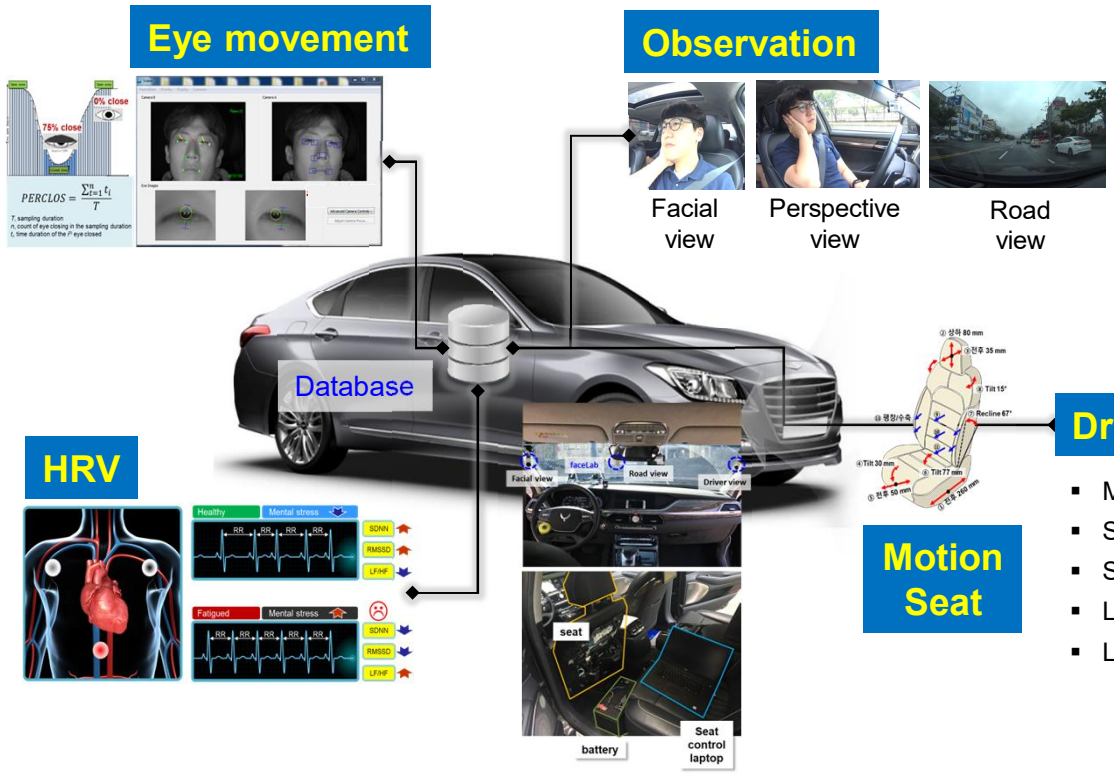
- A **motion seat system** can contribute to **prevention of a car accident in autonomous driving** by reducing the take-over time through helping the driver maintain his/her alertness level



Follow-Up Research

- Verified the effectiveness of the motion seat system by in-vehicle testing for reduction of passive TR fatigue

In-vehicle testing



- Mean velocity
- SD of velocity
- Steering wheel rate
- Lateral acceleration
- Longitudinal acceleration

sensors MDPI

Article

Effects of a Motion Seat System on Driver's Passive Task-Related Fatigue: An On-Road Driving Study

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check for updates

Abstract: Passive task-related (TR) fatigue caused by monotonous driving can negatively affect driving safety by impairing driver alertness and performance. This study aims to evaluate the effectiveness of a motion seat system on the driver's passive TR fatigue in terms of driving performance, physiological response, and subjective fatigue by using automotive and physiological sensors those applicable to on-road driving environment. Twenty drivers (5 females and 15 males; age = 38.5 ± 12.2) with more than two years of driving experience participated in an on-road experiment with two driving conditions: driving in the static seat condition during the first half of the driving session and then in the static (static–static, SS) or motion seat (static–motion, SM) condition during the second half. The SM condition showed significantly lower passive TR fatigue by 4.4~56.5% compared to the SS condition in terms of the standard deviation of velocity, percentage of eyelid closure rate (PERCLOS), and the ratio of low- to high-frequency power (LF/HF) of electrocardiography signals. The drivers rated significantly lower subjective state changes of overall fatigue, mental fatigue, passive TR fatigue, drowsiness, and decreased concentration in the SM condition than those in the SS condition. The findings of the study support the use of a motion seat system can be an effective countermeasure to reduce passive TR fatigue.

Keywords: passive task-related driver fatigue; monotonous driving; fatigue countermeasure; motion seat system

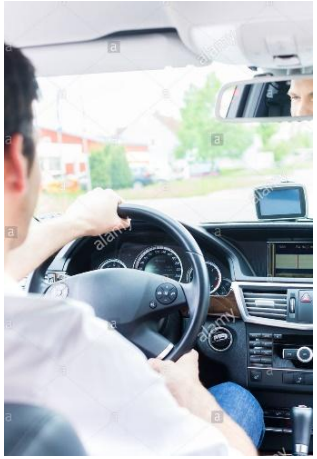
Commercialization of Motion Seat System

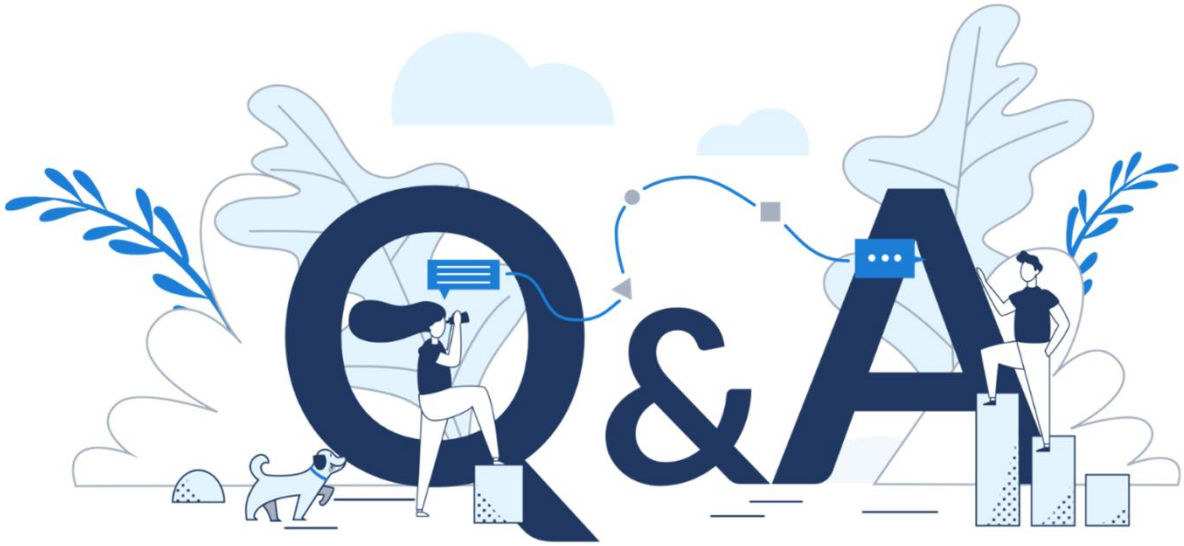
G80 and GV80 | Ergo Motion Seat | Genesis - YouTube

Commercialized in Dec. 2020

Future Research

- Analyze the **effect of a motion seat system on passive TR fatigue** by other factors including **driving experience, age, and fatigue propensity**
  - ✓ The cognitive workload of a driver can vary by driving experience (novice vs. experienced), gender, and age
  - ✓ The degree of passive TR fatigue can vary by individual fatigue propensity, which can be identified by a driver stress inventory and/or a driving coping questionnaire



Q & A

Thank you for your attention!